

Nikolai Goncharov

PhD Candidate, Australian Centre for Robotics
University of Sydney, Australia
☎ +61 457 264 001

✉ nikolai.goncharov@sydney.edu.au
in linkedin
g github

EDUCATION

University of Sydney

Nov 2022 – Jun 2026

PhD in Engineering

Thesis: “Robust Object Tracking under View-Dependent Appearance using Light Fields”

Supervisors: Dr. Donald Dansereau, Dr. Stewart Worrall

Submission Date: June 30, 2026

Skolkovo Institute of Science and Technology

2020 – 2022

Master of Science in Computer Science. GPA: 3.6/4

Thesis: “Segmentation of Planes in Point Clouds with 3D Neural Networks”

Supervisor: Gonzalo Ferrer

Moscow Institute of Electronics and Mathematics

2016 – 2020

Bachelor of Science in Computer Engineering. GPA: 3.7/4

RESEARCH INTERESTS

I am currently developing light-based scene representations to enable robust perception in the presence of reflective and refractive objects.

Computer vision, complex visual appearance in dynamic scenes, learnable rendering representations, reflectance modeling, light field imaging, physically-based rendering, 3D object tracking.

PUBLISHED WORKS

Segment Anything in Light Fields for Real-Time Applications

Nikolai Goncharov, Donald G. Dansereau

WACV workshop, 2025

EVOPS Benchmark: Evaluation of Plane Segmentation from RGBD and LiDAR Data

Anastasiia Kornilova, Dmitrii Iarosh, Denis Kukushkin, **Nikolai Goncharov**, Pavel Mokeev, Arthur Saliou, Dr. Gonzalo Ferrer

IROS, 2022

WORKS UNDER REVIEW

Light Field Based 6DoF Tracking of Previously Unobserved Objects

Nikolai Goncharov, James L. Gray, Donald G. Dansereau

ICRA workshop, 2026

A 3D Reconstruction Benchmark for Asset Inspection

James L. Gray, **Nikolai Goncharov**, Alexandre Cardaillac, Ryan Griffiths, Jack Naylor, Donald G. Dansereau

ECCV, 2026

WORKS IN PREPARATION

Robust Object Tracking under View-Dependent Appearance using Light Fields

Nikolai Goncharov, Donald G. Dansereau

Planned venue: *ACCV*, 2026

INDUSTRY EXPERIENCE

Yandex Eats, Research Engineer

Sep 2021 – Mar 2023

I gained my industry experience developing machine learning models for recommender systems.

Yandex Self-Driving Cars, Research Intern

Jun 2021 – Sep 2021

During this internship, I worked on LiDAR point clouds segmentation in autonomous driving under challenging sensing conditions.

AWARDS AND SCHOLARSHIPS

NVIDIA Academic Grant

2025

Physics-Based Simulation for Robotic Vision in Extreme Environments (AUD\$48k, group grant)

University of Sydney

2023

Faculty of Engineering Research Scholarship

Skolkovo Institute of Science and Technology

2020

Masters Scholarship

SERVICE

Seminar Series, Australian Centre for Robotics

2025

Speaker. *Video link*

Robotic Vision Summer School

2025

Organizers Committee Member, Tutor

Robotic Imaging Lab

2024 – Now

Lab Manager

Faculty of Engineering

2024 – Now

Mentor. Undergraduate students: *Antonio Matrugilo*, *Daniel Biswas*

Robotic Imaging Reading Group

2023 – Now

Speaker

TEACHING

University of Sydney

2025

Computer Vision, Intro to Programming, Intro to Engineering Computing

University of Sydney

2023 – 2024

Mechatronics System Design